

BILKENT UNIVERSITY

Engineering Mathematics I Midterm 1				
Code : Math 241	Last Name: _____			
Acad. Year: 2019-2020	Name : _____		Student No.: _____	
Semester : Fall	Section: _____		Signature: _____	
Date : 25.11.2019	4 QUESTIONS ON 4 PAGES TOTAL 100 POINTS			
Time : 17:40				
Duration : 120 minutes				
1	2	3	4	

No partial credit will be given for unsupported answers!

1. (25 pts) Let $V = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : v_1 \in \mathbb{R}_+, v_2 \in \mathbb{R}_+ \right\}$, where $\mathbb{R}_+$ denotes the set of positive real numbers. For any $v, u \in V$ and $\alpha \in \mathbb{R}$, let the addition and scalar multiplication be defined by $v + u = \begin{bmatrix} v_1 u_1 \\ v_2 u_2 \end{bmatrix}$; $\alpha v = \begin{bmatrix} v_1^\alpha \\ v_2^\alpha \end{bmatrix}$.

(a) Show that V is a vector space over $\mathbb{R}$.

(b) Find a basis and dimension of V .

(a) (i) $v + u = \begin{bmatrix} v_1 u_1 \\ v_2 u_2 \end{bmatrix} = \begin{bmatrix} u_1 v_1 \\ u_2 v_2 \end{bmatrix} = u + v, \quad \forall u, v \in V$

(ii) $(v + u) + w = \begin{bmatrix} v_1 u_1 \\ v_2 u_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} (v_1 u_1) w_1 \\ (v_2 u_2) w_2 \end{bmatrix} = \begin{bmatrix} v_1 (u_1 w_1) \\ v_2 (u_2 w_2) \end{bmatrix}$
 $= \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} u_1 w_1 \\ u_2 w_2 \end{bmatrix} = v + (u + w), \quad \forall u, v, w \in V$

(iii) $0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in V$ and $0 + v = \begin{bmatrix} 1 v_1 \\ 1 v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = v \Rightarrow 0$ is additive identity.

(iv) $v + 0 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in V, \quad 1 \cdot v = \begin{bmatrix} 1 v_1 \\ 1 v_2 \end{bmatrix} \in V$ and $v + (1 \cdot v) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0$

(v) $\forall \alpha \in \mathbb{R}, \quad \alpha(v + u) = \alpha \begin{bmatrix} v_1 u_1 \\ v_2 u_2 \end{bmatrix} = \begin{bmatrix} v_1^\alpha u_1^\alpha \\ v_2^\alpha u_2^\alpha \end{bmatrix} = \begin{bmatrix} v_1^\alpha \\ v_2^\alpha \end{bmatrix} + \begin{bmatrix} u_1^\alpha \\ u_2^\alpha \end{bmatrix}$
 $= \alpha v + \alpha u$

(vi) $\forall \alpha, \beta \in \mathbb{R}, \quad (\alpha + \beta)u = \begin{bmatrix} u_1^{\alpha+\beta} \\ u_2^{\alpha+\beta} \end{bmatrix} = \begin{bmatrix} u_1^\alpha u_1^\beta \\ u_2^\alpha u_2^\beta \end{bmatrix} = \begin{bmatrix} u_1^\alpha \\ u_2^\alpha \end{bmatrix} + \begin{bmatrix} u_1^\beta \\ u_2^\beta \end{bmatrix}$
 $= \alpha u + \beta u$

$$(vii) \forall \alpha, \beta \in \mathbb{R} \quad (\alpha\beta)u = \begin{bmatrix} u_1 \alpha\beta \\ u_2 \alpha\beta \end{bmatrix} = \begin{bmatrix} (u_1\alpha)\beta \\ (u_2\alpha)\beta \end{bmatrix} = \beta \begin{bmatrix} u_1\alpha \\ u_2\alpha \end{bmatrix}$$

$$= \beta(\alpha u)$$

$$(viii) \forall u \in V, \quad 1u = 1 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1 \cdot 1 \\ u_2 \cdot 1 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = u$$

$\Rightarrow V$ is a vector space over $\mathbb{R}$.

(b) Claim: $\left\{ \begin{bmatrix} u_1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ u_2 \end{bmatrix} \right\}$ is a basis for V for any $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Pf of claim $\forall u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in V$, let $\alpha = \log_{u_1} u_1$
 $\beta = \log_{u_2} u_2$

$$\text{Then } \alpha \begin{bmatrix} u_1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1^{\log_{u_1} u_1} \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ u_2^{\log_{u_2} u_2} \end{bmatrix}$$

$$= \begin{bmatrix} u_1 \cdot 1 \\ 1 \cdot u_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = u$$

$\Rightarrow \left\{ \begin{bmatrix} u_1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ u_2 \end{bmatrix} \right\}$ spans V .

$$\alpha \begin{bmatrix} u_1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{matrix} u_1^\alpha = 1 \Rightarrow \alpha = 0 \\ u_2^\beta = 1 \Rightarrow \beta = 0 \end{matrix}$$

$\Rightarrow \left\{ \begin{bmatrix} u_1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ u_2 \end{bmatrix} \right\}$ is L.I. $\Rightarrow \left\{ \begin{bmatrix} u_1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ u_2 \end{bmatrix} \right\}$ is a basis for V .

$$\Rightarrow \dim V = 2.$$

2. (25 pts) Let

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

(a) Find the eigenvalues and corresponding eigenspaces of A .

(b) If A is diagonalizable, find a diagonalizing matrix Q such that $Q^{-1}AQ = D$ is diagonal. If not, explain why it is not diagonalizable.

(a) char poly $\begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda)^2 = 0$

$\Rightarrow \lambda_1 = \lambda_2 = 1$ $\lambda_3 = 2$ are the eigenvalues.

To find eigenspace corr to $\lambda = 1$

$$\begin{bmatrix} 0 & 0 & -1 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{bmatrix} \xrightarrow{\text{e.r.o.}} \begin{bmatrix} 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 = \alpha \\ x_2 = \beta \\ x_3 = 0 \end{matrix}$$

$W_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is the corresponding eigenspace.

eigenspace corresponding to $\lambda = 2$

$$\begin{bmatrix} -1 & 0 & -1 & | & 0 \\ 0 & -1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{\text{e.r.o.}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{matrix} x_3 = \alpha \\ x_2 = \alpha \\ x_1 = -\alpha \end{matrix}$$

$W_2 = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$ is the eigenspace for $\lambda = 2$.

(b) since A has 3 L.I. eigenvectors, it is diagonalizable
and $Q = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ diagonalizes A and $Q^{-1}AQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

3. (25 pts) Fit a quadratic polynomial to the data set

$$\{(-2, 6), (-1, 4), (0, 2), (1, 0)\}$$

using least squares approximation method.

We want to fit $f(x, \theta) = \theta_0 + \theta_1 x + \theta_2 x^2$ to

-2	6
-1	4
0	2
1	0

$\Rightarrow$ Normal eqns $X^T X \theta = X^T y$ must be satisfied with

$$X = \begin{bmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 \\ 4 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 4 \\ -1 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 \\ 4 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & -2 & 6 \\ -2 & 6 & -8 \\ 6 & -8 & 18 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 14 \\ -14 \\ 30 \end{bmatrix}$$

To find $\theta_0, \theta_1, \theta_2$, we will use r.r.o.

$$\left[\begin{array}{ccc|c} 4 & -2 & 6 & 14 \\ -2 & 6 & -8 & -14 \\ 6 & -8 & 18 & 30 \end{array} \right] \xrightarrow{\substack{2R_2 + R_1 \rightarrow R_1 \\ 3R_2 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 0 & 10 & -10 & -14 \\ -2 & 6 & -8 & -14 \\ 0 & 10 & -6 & -12 \end{array} \right] \xrightarrow{\substack{-1/2 R_2 \rightarrow R_2 \\ -R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 0 & 10 & -10 & -14 \\ 1 & -3 & 4 & 7 \\ 0 & 0 & 4 & 2 \end{array} \right]$$

$$\xrightarrow{\substack{R_1 \leftrightarrow R_2 \\ R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & -3 & 4 & 7 \\ 0 & 10 & -10 & -14 \\ 0 & 0 & 4 & 2 \end{array} \right] \xrightarrow{\substack{R_1 \rightarrow R_1 \\ R_2 \rightarrow R_2}} \left[\begin{array}{ccc|c} 1 & -3 & 4 & 7 \\ 0 & 1 & -1 & -1.4 \\ 0 & 0 & 1 & 0.5 \end{array} \right] \Rightarrow \begin{aligned} \theta_2 &= 112 \\ \theta_1 &= -9/10 \\ \theta_0 &= 23/10 \end{aligned}$$

$$\Rightarrow f(x, \theta) = \frac{23}{10} - \frac{9}{10}x + \frac{x^2}{2}$$

4. (25 pts) Let A be a matrix of order n . Prove the following:

(a) If λ is a nonzero eigenvalue of A , then $\frac{\det(A)}{\lambda}$ is an eigenvalue of $\text{Adj } A$.

(b) If v is an eigenvector of A , then v is also an eigenvector of $\text{Adj } A$.

(a) Let $\lambda \neq 0$ be an eigenvalue of A . Then $|A - \lambda I| = 0$. On the other hand, $A \cdot \text{Adj } A = \det A \cdot I$ and

$$\begin{aligned} \left| \text{Adj } A - \frac{\det A}{\lambda} I \right| &= \left| \text{Adj } A - \frac{A \cdot \text{Adj } A}{\lambda} \right| \\ &= \left| \frac{1}{\lambda} \text{Adj } A (\lambda I - A) \right| = \left(\frac{1}{\lambda} \right)^n |\text{Adj } A| \underbrace{|\lambda I - A|}_{=0} \\ &= 0. \end{aligned}$$

$\Rightarrow \frac{\det A}{\lambda}$ is an eigenvalue of $\text{Adj } A$.

(b) Let v be an eigenvector of A corresponding to the eigenvalue λ , i.e., $Av = \lambda v$. Then consider

$$\begin{aligned} \left(\text{Adj } A - \frac{\det A}{\lambda} I \right) v &= \frac{1}{\lambda} (\lambda \text{Adj } A - \det A \cdot I) v \\ &= \frac{1}{\lambda} (\underbrace{\lambda \text{Adj } A}_{\uparrow} v - \det A \cdot v) \\ &= \frac{1}{\lambda} (\underbrace{\text{Adj } A}_{\uparrow} (Av) - \det A \cdot v) \\ &= \frac{1}{\lambda} (\det A \cdot I v - \det A \cdot v) \\ &= 0 \Rightarrow v \text{ is an eigenvector.} \end{aligned}$$